

SMART QUIZ MAKER APP - Project Report

- Project Name: **SMART QUIZ MAKER APP**

- Member's Names & Roll no.:

1. _____
2. _____
3. _____
4. _____

- Course / Semester: _____

- Supervisor / Teacher Name:

- Date: _____

Abstracts:

- This project is a console-based application that allows teachers to create, manage, and preview quizzes, and students to take quizzes and receive automatic scores.
 - The system includes user authentication for Admin, Teacher, and Student roles, with persistent file-based storage.
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Introduction:

The SMART QUIZ MAKER APP is designed to simplify the quiz creation and evaluation process. It enables teachers to create quizzes with multiple-choice questions, allows students to take timed quizzes, and stores scores for review. The system uses C++ programming language with file handling for storing data.

Software Requirements:

- IDE/Compiler: Dev-C++
 - Programming Language: C++
 - Libraries Used:
 - → Input/Output
 - → File handling
 - → String manipulation
 - → Dynamic arrays
 - → Randomization
 - → swap(), random_shuffle()
 - → Timer for quizzes
-

System Design / Architecture:

- **User Roles:**

1. Admin: Manage users, generate credentials
2. Teacher: Create/Edit quizzes, preview quizzes, view results
3. Student: Take quizzes, view scores, update profile

- **File Structure:**

- adminFile.txt → Admin credentials
- teacherFile.txt → Teacher info
- studentFile.txt → Student info
- quizA.txt / quizB.txt → Questions for sections
- scoreA.txt / scoreB.txt → Student scores

- **Data Structures:**

- struct Quiz → Quiz metadata
- struct Question → Question statement, options, correct answer
- struct Details → User information
- vector → Stores all questions in memory

Key Features / Highlights:

- User authentication (Admin / Teacher / Student)
- Create quizzes with any number of questions
- Shuffle questions and options for fairness
- Automatic scoring for students
- File-based storage (persistent data)
- Stopwatch feature for timed quizzes
- Simple console interface (easy to use)

Modules / Functions:

Function Name	Purpose
<code>myMain()</code>	Main menu of the app
<code>adminDashboard()</code>	Admin login and features
<code>teacherDashboard()</code>	Teacher login and features
<code>studentDashboard()</code>	Student login and features
<code>createQuiz()</code>	Teacher creates quiz and saves in file
<code>editQuiz()</code>	Teacher edits existing quiz
<code>readQuiz()</code>	Student takes quiz; teacher previews quiz
<code>viewResult()</code>	Admin/teacher views student scores
<code>shuffleOptions()</code>	Randomizes options of a question
<code>random_shuffle(questions.begin(), questions.end())</code>	Randomizes question order

Code Snippets / Explanation:

- **`shuffleOptions()`**: Randomizes options and updates the correct option
- **`random_shuffle()`**: Randomizes the order of questions
- **File reading/writing**: Stores quizzes and scores for persistence

Testing:

- **Example:**
 - Create a quiz with 2 questions
 - Student takes quiz
 - System calculates and stores score

```
C:\Users\Mohammad bin Sajjad\Desktop\PF Project\main.exe
Choose option:
1. Update Info
2. Take Quiz
0. Logout
2
Which Section's quiz do you want to take? (A or B) : A

Question 1: 1+3
1) 6
2) 4
3) 5
4) 7
Enter correct option (1-4) : 2

Correct!!
Your score is : 1
Question 2: 3+2 = ?
1) 5
2) 7
3) 4
4) 2
Enter correct option (1-4) : 2

Incorrect !!
Time Taken: 0 minutes
You choose Student Dashboard
<===== Welcome to Student Dashboard =====>

Enter Student Username: _
```

Main Menu Console Screen:

```
C:\Users\Mohammad bin Sajjad\Desktop\PF Project\main.exe
<===== Welcome to our SMART QUIZ MAKER APP =====>

Choose which feature do you want to use?
1. Admin Feature
2. Student Feature
3. Teacher Feature
0. Exit
```

Student Dashboard Console Screen:

```
Enter Student Username: student
Enter Student Password: 1234

Choose option:
1. Update Info
2. Take Quiz
0. Logout
```

Teacher Dashboard Console Screen:

```
Enter Teacher Username: Teacher
Enter Teacher Password: 1234

Choose option:
1. Update Info
2. Create Quiz
3. Edit Quiz
4. View Result
5. Preview Quiz
0. Logout
```

Conclusion:

The SMART QUIZ MAKER APP successfully implements user management, quiz creation, question randomization, timed quizzes, and score calculation. Future improvements could include a GUI interface, more question types, and an online version.

References / Acknowledgements:

- Websites => Geeks for Geeks , MDN Docs.
- Tutorials => Generate Random Number , Timmer Func.